Week 9

Write a Go program to calculate average using arrays.

Code

```
package main

import "fmt"

func main() {
    a := [5]int{10, 20, 30, 40, 50}
    sum := 0

    for _, n := range a {
        sum += n
    }

    avg := float64(sum) / float64(len(a))
    fmt.Println("Average =", avg)
}
```

Output

```
Average = 30
```

Week 10

Write a Go program to delete duplicate elements in a given array.

Code

```
package main

import "fmt"

func main() {
    arr := []int{2, 4, 1, 2, 1}

    for i := 0; i < len(arr); i++ {
        for j := i + 1; j < len(arr); j++ {
            if arr[i] == arr[j] {
                arr[j] = 0
            }
        }
    }

    for _, n := range arr {
        if n != 0 {
            fmt.Print(n, " ")
        }
    }
}
```

Output

```
2 4 1
```

Week 11

Write a Go program with examples of array reverse/sort functions for integers and strings.

Code

```
package main

import (
    "fmt"
    "slices"
)

func main() {
    a := []int{1, 2, 3, 4}
    b := []string{"Tan", "Veer", "Rao"}

    slices.Sort(a)
    fmt.Println("Sorted integers:", a)

    slices.Reverse(a)
    fmt.Println("Reverse integers:", a)

    slices.Sort(b)
    fmt.Println("Sorted strings:", b)

    slices.Reverse(b)
    fmt.Println("Reverse strings:", b)
}
```

Output

```
Sorted integers: [1 2 3 4]
Reverse integers: [4 3 2 1]
Sorted strings: [Rao Tan Veer]
Reverse strings: [Veer Tan Rao]
```

Week 12

Write a program demonstrating the strings Contains, ContainsAny, Count, and EqualFold functions.

Code

```
package main

import (
    "fmt"
    "strings"
)

func main() {
    s := "Hello World"

    fmt.Println(strings.Contains(s, "Hello"))
    fmt.Println(strings.ContainsAny(s, "aeiou"))
    fmt.Println(strings.Count(s, "l"))
    fmt.Println(strings.EqualFold("hello", "HELLO"))
}
```

Output

```
true
true
3
true
```